The Political Economy of Pandemic Restrictions: A Data-Driven Analysis of State Repression During COVID-19

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Abstract

This project investigates the relationship between COVID-19 government restrictions ("lockdowns") and state repression events in active conflict zones from 2020 to 2023. Using data from the Oxford COVID-19 Government Response Tracker (OxCGRT) and the Armed Conflict Location & Event Data Project (ACLED), we analyze whether governments utilized pandemic measures as a tool for political repression and whether economic support mitigated potential violence. Our findings confirm a significant positive correlation between stringency and repression ("The Stick") and a mitigating effect of economic support ("The Carrot"). Machine learning models reveal a structural linear relationship where high restrictions without adequate economic aid often precipitate extreme violence, identifying a specific "danger zone" for fragile states.

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1. Introduction

1.1 Motivation and Personal Context

The COVID-19 pandemic catalyzed a global shift in governance, granting states unprecedented emergency powers. As a student during this era, I witnessed how lockdowns fundamentally altered social fabrics. Beyond the public health utility, these restrictions raised profound questions about power dynamics: Were these measures applied solely for health, or did they serve ulterior political motives? This project seeks to empirically test the intuition that emergency powers can be weaponized against citizens.

1.2 Research Question

> *"Did governments use the power granted by COVID-19 restrictions as a political tool to repress their own citizens, and did economic support mitigate this repression?"*

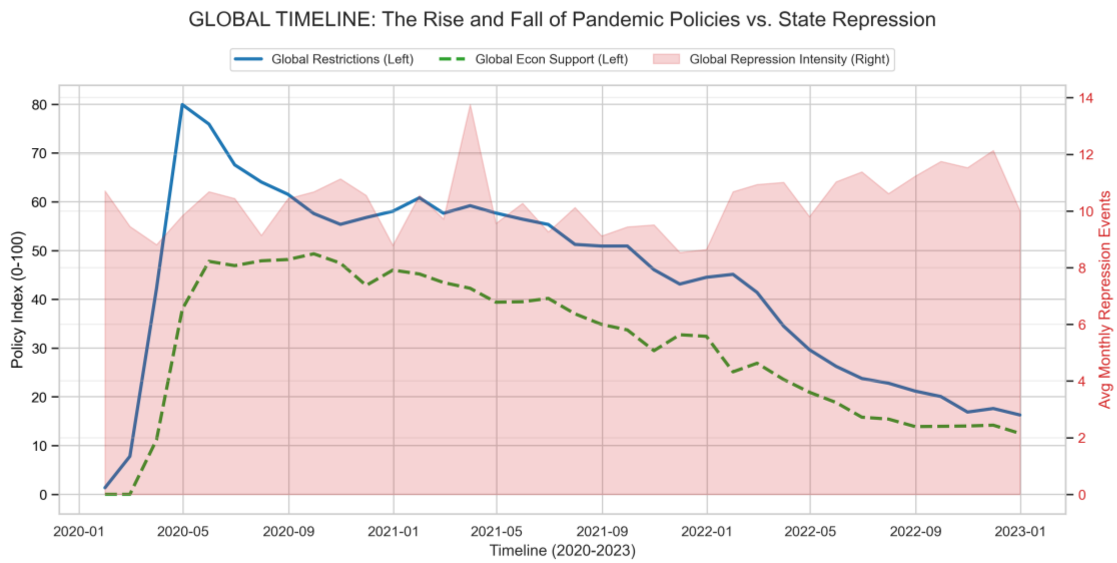

Figure 1: Timeline of COVID-19 Stringency and Repression Events (2020-2023).

To answer this, we examine the interplay between three core variables: * **Government Stringency** (Lockdowns, bans) * **Economic Support** (Stimulus, debt relief) * **State Repression** (Violence against civilians)

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2. Data and Methodology

2.1 Data Sources

We integrated two primary datasets to construct a comprehensive view of the pandemic's political landscape:

1. **OxCGRT (Oxford COVID-19 Government Response Tracker)**:
   - *Features*: `StringencyIndex_Average` (0-100), `EconomicSupportIndex` (0-100).
   - *Scope*: National-level daily policy data (2020-2023).
   - *Source*: Blavatnik School of Government, University of Oxford.
2. **ACLED (Armed Conflict Location & Event Data Project)**:
   - *Target Variable*: `RepressionEvents` (Count of state Violence Against Civilians).
   - *Context Variables*: `PoliticalViolenceEvents`, `AnnualDemonstrations`.
   - *Source*: ACLED Global Data.

2.2 Preprocessing

- **Temporal Aggregation**: Daily OxCGRT data was downsampled to monthly averages to match ACLED's granularity.
- **Merging**: Datasets were joined on `Country` and `Month`, resulting in a unified time-series panel for 85+ countries.

- **Lag Generation**: Time-lagged variables (t-1 to t-12) were created to test delayed effects of policy implementation.

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3. Results: Statistical Analysis

3.1 Hypothesis 1: The "Stick" Effect (Confirmed)

We hypothesized that higher restrictions lead to increased repression. * **Finding**: A statistically significant positive correlation exists between `StringencyIndex` and `RepressionEvents` (Spearman $\rho > 0, p < 0.05$). * **Regression**: The Linear Regression coefficient for Stringency is **+0.017**, indicating that for every point increase in stringency, repression events rise, holding other factors constant.

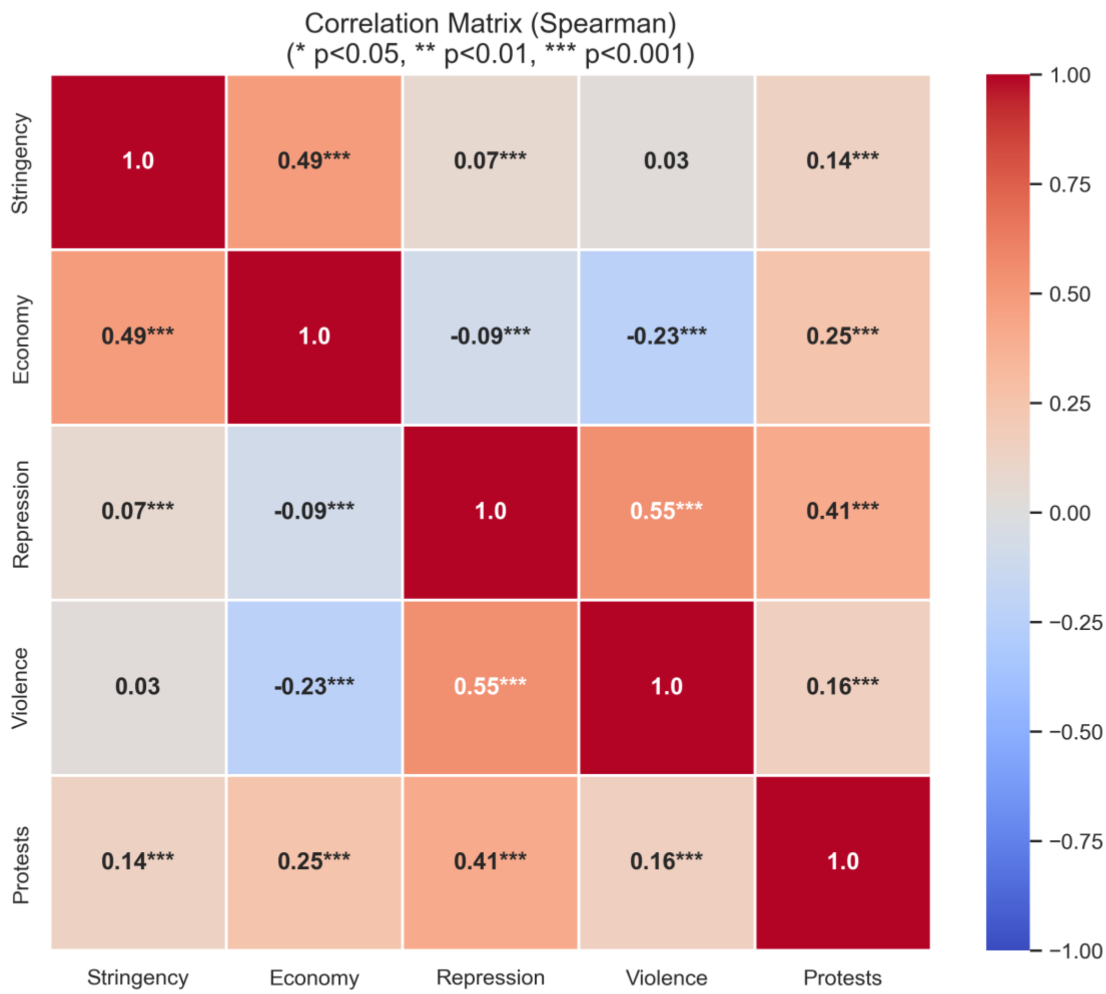

Figure 2: Correlation Matrix highlighting the relationship between Stringency Index and Repression Events.

3.2 Hypothesis 2: The "Delayed Effect" (Rejected)

We tested if repression peaks months after restrictions are announced. * **Finding**: The strongest correlation was observed at **Lag 0** (same month). * **Interpretation**: This supports the **"Immediate Enforcement Hypothesis."** Police crackdowns and state violence occur simultaneously with the announcement of bans, rather than as a delayed reaction.

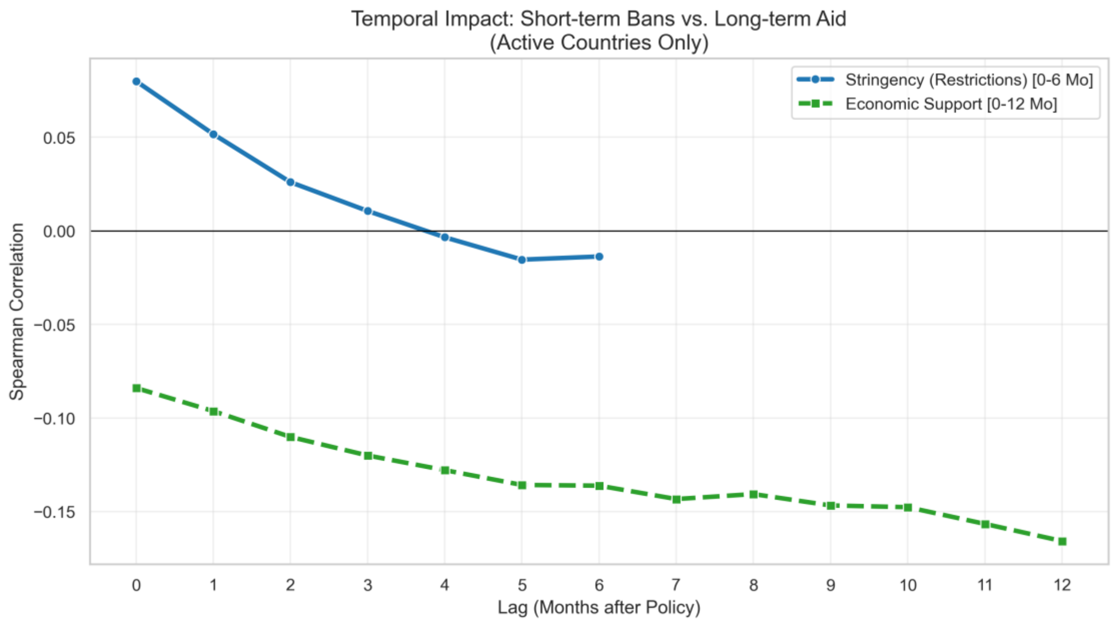

Figure 3: Temporal Lag Analysis showing the correlation strength peaking at t=0 (Same Month).

3.3 Hypothesis 3: The "Carrot" Effect (Confirmed)

We hypothesized that economic support mitigates violence. * **Finding**: `EconomicSupportIndex` shows a negative coefficient (**-0.010**) in regression models. * **Insight**: Economic aid acts as a pacifier. However, T-tests reveal this effect is conditional: it works best under low-to-moderate restrictions. Under extreme authoritarianism (high stringency), economic aid is insufficient to curb violence.

3.4 Temporal Evolution of the Relationship

The relationship between restrictions and repression was not static. As the pandemic progressed, the correlation evolved.

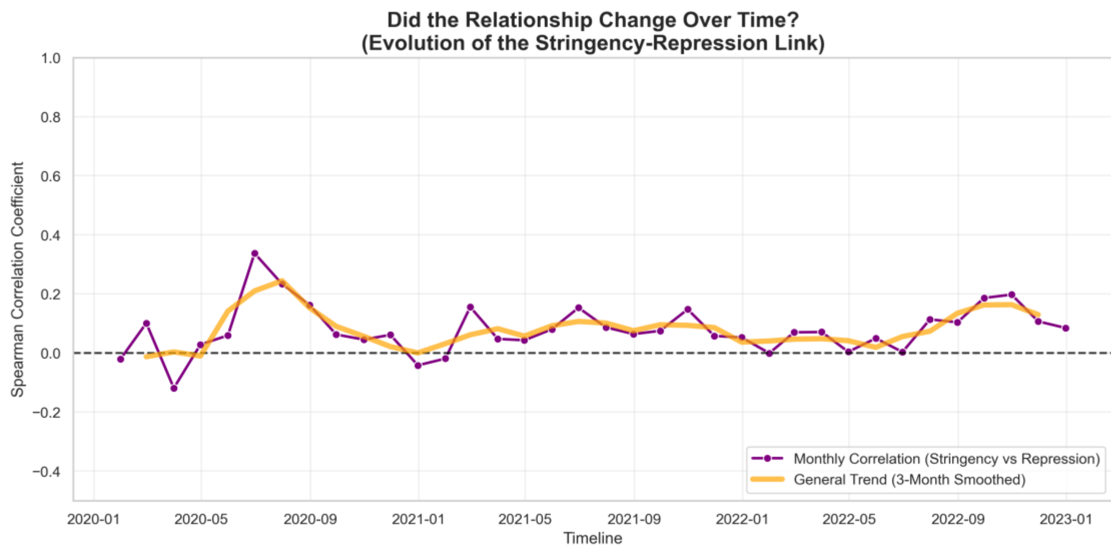

Figure 4: Rolling correlation between Stringency and Repression over time, showing peaks during major restriction waves.

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4. Results: Machine Learning

To validate the statistical findings, we employed supervised and unsupervised learning techniques.

4.1 Predictive Modeling

We trained models to predict next month's repression events.

Model	Test R^2 Score	MAE	Key Takeaway
Linear Regression	**0.73**	4.50	The relationship is structural and linear.
Random Forest	0.56	4.63	Non-linear complexity implies overfitting; the linear signal is stronger.

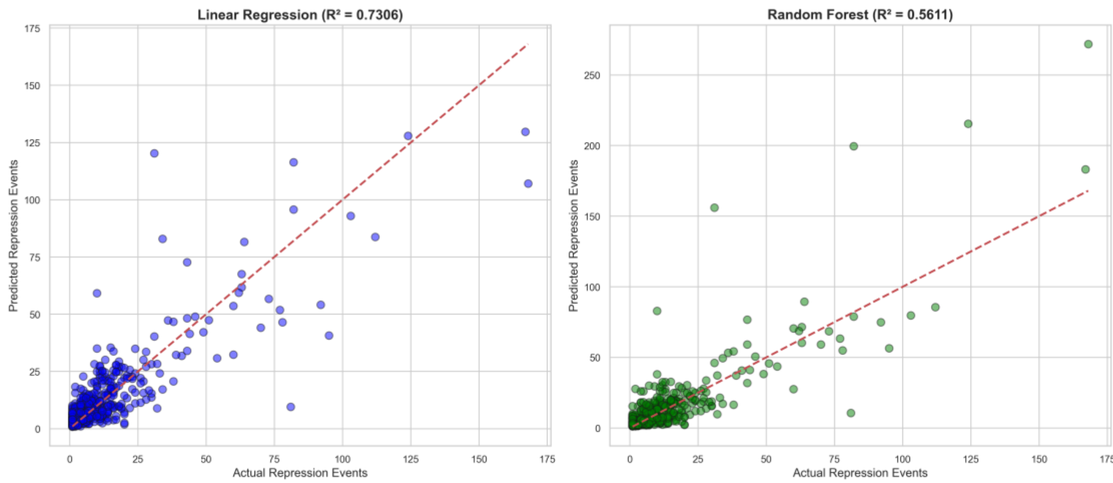

Figure 5: Scatter plot comparing Actual vs Predicted values for Linear Regression and Random Forest.

Feature Importance: Both models identified **Past Repression** as the strongest predictor (Momentum Effect), followed immediately by **Stringency Index**, validating the importance of policy decisions in driving future violence.

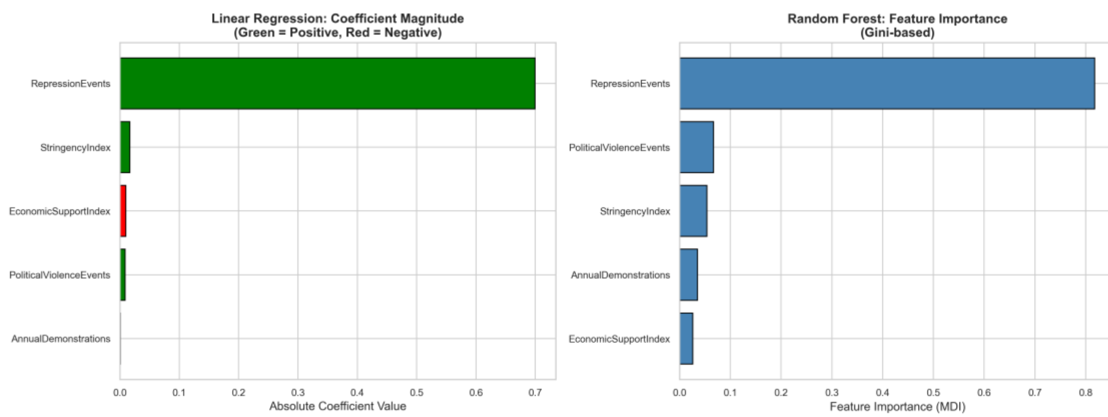

Figure 6: Comparison of Feature Importance in Random Forest vs Linear Regression models.

4.2 Clustering: Archetypes of State Behavior

Using K-Means Clustering ($k = 3$), we identified three distinct country profiles:

1. **The Welfare State (Cluster 1)**:
   - *Characteristics*: High Restrictions (84/100), High Economic Support (62/100).
   - *Outcome*: **Low Repression** (~7 events/month).
   - *Interpretation*: Wealthy nations successfully "bought" social peace.
2. **The Authoritarian Outlier (Cluster 2 - e.g., Myanmar, India)**:
   - *Characteristics*: Moderate Restrictions, Moderate Support.
   - *Outcome*: **Extreme Repression** (~116 events/month).

- *Interpretation*: Fragile states with pre-existing conflicts used restrictions to amplify control, leading to violence **16x higher** than welfare states.

3. **The Passive State (Cluster 0)**:
   - *Characteristics*: Low Restrictions, Low Support.
   - *Outcome*: Low Repression.

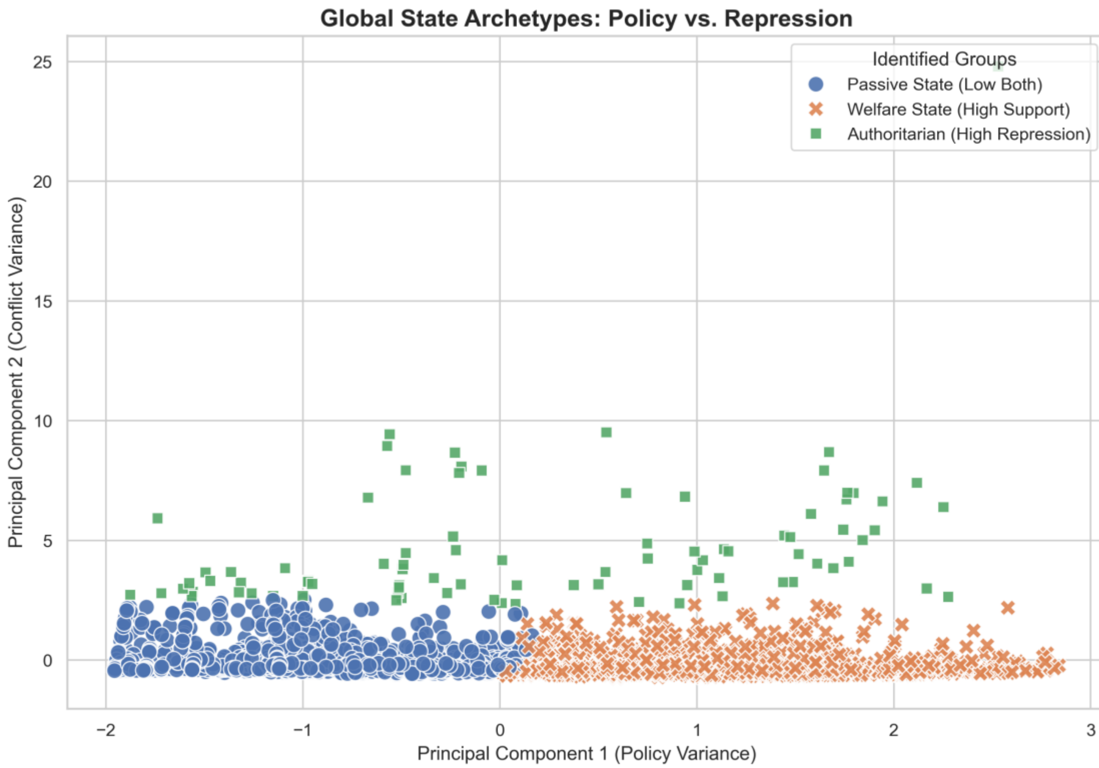

Figure 7: K-Means Clustering identifying distinct state archetypes based on policy responses.

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5. Discussion

5.1 The "Inequality Trap"

The most profound finding is the disparity between clusters. Stability during the pandemic was a luxury. Developed nations (Welfare States) could afford to balance strict lockdowns with generous economic aid ("The Carrot"), maintaining order. Developing or authoritarian states, lacking fiscal capacity, relied solely on "The Stick," creating a cycle of enforcement and violence.

5.2 Validity of the Linear Model

The superiority of Linear Regression (R^2=0.73) over Random Forest suggests that the mechanics of state repression are not mysterious or highly complex. They follow a predictable, linear path: **Pressure (Restrictions) - Relief (Aid) = Violence.**

6. Conclusion

This project successfully answers the research question affirmative: **Yes, COVID-19 restrictions were significantly correlated with state repression**, particularly in contexts where economic support was lacking.

The data confirms a “Carrot & Stick” dynamic: * **The Stick**: Lockdowns provided the legal framework and immediate trigger for increased state violence. * **The Carrot**: Economic support effectively mitigated this violence, but only for nations that could afford it.

Ultimately, the pandemic exposed a fragility in the global social contract. Emergency powers, once granted, are difficult to retract. This study serves as a quantitative warning: without the buffer of economic security, public health mandates readily devolve into instruments of political repression. The mechanism is not merely “unrest leading to crackdown,” but rather a structural feature where high stringency gives the state the legal and physical tools to dominate the public sphere, often shifting the balance of power permanently in favor of the executive branch.

As we move beyond the acute phase of the pandemic, the legacy of these policies remains. The “Stick” has been normalized in many fragile democracies, while the “Carrot” has evaporated with the end of fiscal stimulus. This leaves a dangerous vacuum where state capacity for violence remains elevated, but its capacity for care has diminished.

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7. Limitations and Future Work

- **Complex Cause-and-Effect Relationships**: While Lag analysis supports the directionality (Restrictions → Repression), establishing pure cause-and-effect remains a challenge. Governments might anticipate unrest and impose restrictions preemptively. Unobserved variables, such as internal military coup attempts or external geopolitical pressures, could influence both restrictions and repression simultaneously.
- **Temporal Resolution and Granularity**: Our analysis aggregates daily policy data into monthly averages. This inevitably smooths out short-term escalations. A more detailed daily time-series analysis might reveal finer-grained action-reaction cycles (e.g., a protest on Tuesday leading to a ban on Wednesday).
- **Digital vs. Physical Repression**: This study relies on ACLED, which captures *physical* violence. It does not account for “digital repression” (censorship, internet shutdowns, surveillance), which became a primary tool for authoritarian regimes during the pandemic.
- **The “Announcement vs. Delivery” Gap**: The `EconomicSupportIndex` measures announced policies. In many corrupt or inefficient bureaucratic systems, the money promised may never have reached the citizens, meaning the “Carrot” effect we measured might be an upper bound.

7.2 Future Work

- **Post-Pandemic Legislative Legacy**: A legal analysis of whether emergency decrees initiated in 2020 have been codified into permanent law by 2023.
- **Digital Authoritarianism**: Integrating data on internet shutdowns (e.g., from Access Now) to see if digital repression substituted physical violence in "Welfare States."
- **Sentiment Analysis**: Using X data to measure public sentiment *before* repression events occur, providing a leading indicator for the "Stick" effect.

8. Declaration of Generative AI Use

In accordance with the academic integrity policy, I declare the use of **Google Gemini** to assist in the development of the machine learning code and this report.

- **Scope of Assistance**: AI was used for generating initial syntax for scikit-learn models, debugging error messages in Python scripts, and refining the structure of this final report.
- **Prompt Engineering**: Prompts were used to guide the AI in generating hypothesis testing frameworks and interpreting complex statistical outputs (e.g., Lag Analysis).
- **Ownership of Ideas**: The conceptual framework, research questions, and hypotheses are entirely my own. AI tools were leveraged exclusively to assist with technical implementation and code generation.

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9. References

1. **OxCGRT**: Hale, T., et al. (2021). "A global panel database of pandemic policies (Oxford COVID-19 Government Response Tracker)." *Nature Human Behaviour*.
2. **ACLED**: Raleigh, C., et al. (2010). "Introducing ACLED: An Armed Conflict Location and Event Dataset." *Journal of Peace Research*.